

Source: An Introduction to Contemporary Remote Sensing, 1st Edition

ISBN: 9780071740111

Authors: Qihao Weng Ph.D.

8. Digital Image Analysis

8.1. Introduction

Digital image analysis aims at detecting, identifying, measuring, and analyzing features, objects, phenomena, and processes from digital remote sensing images, and uses computers to process digital images. In the context of digital analysis of remotely sensed data, the basic elements of image interpretation, although initially developed based on aerial photographs, should also be applicable to digital images. That is to say, manual interpretation and analysis is still an important method in digital image processing. However, the majority of digital image analysis methods are based on tone or color, which is represented as a digital number (brightness value) in each pixel of the digital image. The development of digital image analysis and interpretation is closely associated with availability of digital remote sensing images, advances in computer software and hardware, and the evolution of satellite systems. It has now become a broad subject in remote sensing, and therefore it is impossible to discuss all, or even the majority of the methods and techniques in this chapter. Instead, the basic principles of digital image processing are introduced, without getting into advanced algorithms that often require strong mathematical and statistical backgrounds. Before main image analyses take place, preprocessing of digital images are often required. Image preprocessing may include detection and restoration of bad lines, geometric rectification or image registration, radiometric calibration and atmospheric correction, and topographic correction. To make interpretation and analysis easier, it may also be necessary to implement various image enhancements, such as contrast, spatial and spectral enhancements. Finally, we discuss thematic information extraction from remotely sensed image data. The focus is placed on image classification, largely applicable to multi-spectral images, and on the unique methods and techniques for hyperspectral images.

8.2. Image Corrections

Geometric and atmospheric corrections are important steps in image preprocessing. Remote sensing images collected from airborne or spaceborne sensors may contain systematic and nonsystematic errors in terms of image geometry. Systematic errors may be caused by scan skew (the ground swath is not normal to the ground track but is slightly skewed, producing across-scan geometric distortion); mirror-scan velocity variance (the mirror scanning rate is not constant across a given scan, producing along-scan geometric distortion); platform velocity (changes in the speed of the platform produces along-track scale distortion); Earth rotation (Earth's rotation results in a shift of the ground swath being scanned, causing along-scan distortion); and the curvature of the Earth's surface. Nonsystematic errors are created when the sensor platform departs from its normal altitude or the terrain increases in elevation, resulting in changes in image scale. In addition, the axis of a sensor system needs to maintain normal to the Earth's surface and the other parallel to the platform's flight direction. If the sensor departs from this attitude, irregular, random geometric distortion may occur. Geometric correction corrects systemic and nonsystematic errors contained in the remote sensing system and during image acquisition (Lo and Yeung, 2002). It commonly involves: (1) digital rectification, a process by which the geometry of an image is made planimetric; and (2) resampling, a process of extrapolating data values into a new grid. During geometric rectification, we need to identify the image coordinates (i.e., row and column) of several clearly discernible points, called ground control points (GCPs), in the distorted image (A1–A4 in Fig. 8.1) and match them to their true positions in ground coordinates (B1–B4 in Fig. 8.1), which may be measured from a map or a previously rectified image. Once GCP pairs are identified, proper mathematical equations are used to transform the coordinates of the distorted image into new coordinates in alignment with their true ground positions. The procedure of resampling is then followed to determine the new pixel values in the new pixel locations of the corrected image based on the original digital pixel values in the uncorrected image. Three methods are frequently used for resampling: (1) nearest neighbor, (2) bilinear, and (3) cubic convolution. The nearest neighbor resampling uses the input cell value closest to the output cell as the assigned value to the output cell (Fig. 8.2). This method is straightforward and easy to compute. Since it does not alter the original data, it may be the most appropriate method if discrimination between land cover or vegetation types or locating the boundaries among the thematic classes is the top priority in a specific application. However, the nearest neighbor method may result in some pixel values being duplicated while others lost.

Figure 8.1 Image registration.

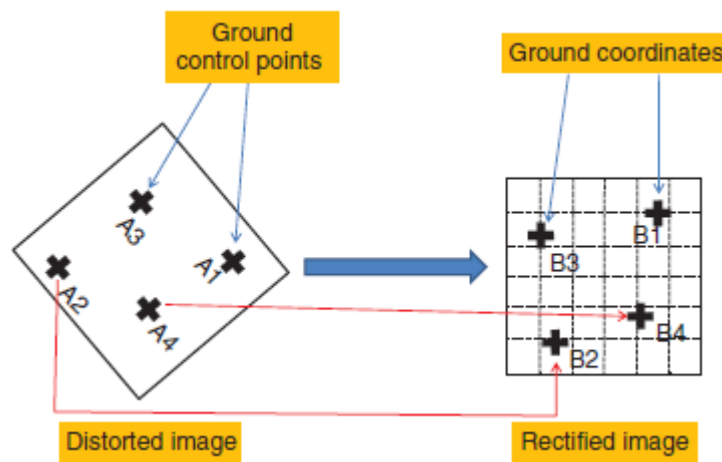
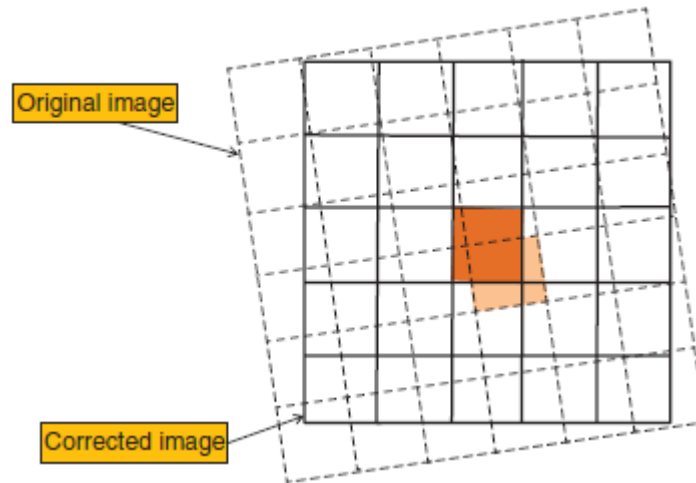


Figure 8.2 The nearest neighbor method of image resampling.



The bilinear interpolation method estimates the output cell value by taking a weighted average of four pixels in the original image nearest to the new cell location (Fig. 8.3). Comparing with the nearest neighbor method, the bilinear interpolation method produces a smoother image, although it is computationally more expensive. Since this method modifies the original data values, it can affect subsequent image analysis. It is recommended to perform bilinear resampling after image classification (Lillesand, Kiefer, and Chipman, 2008). The cubic convolution method estimates the output cell value by calculating the weighted average of the closest sixteen input cells to the new cell location (Fig. 8.4). This method produces the smoothest image, but it requires the most computational resource and time.

Figure 8.3 The bilinear interpolation method of image resampling.

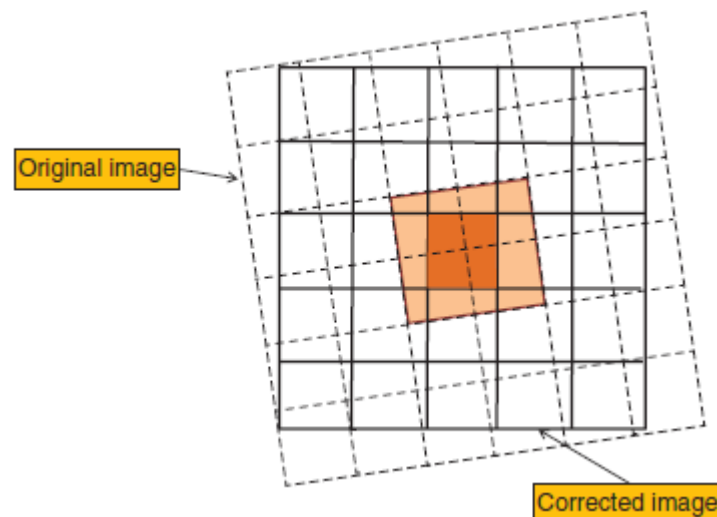
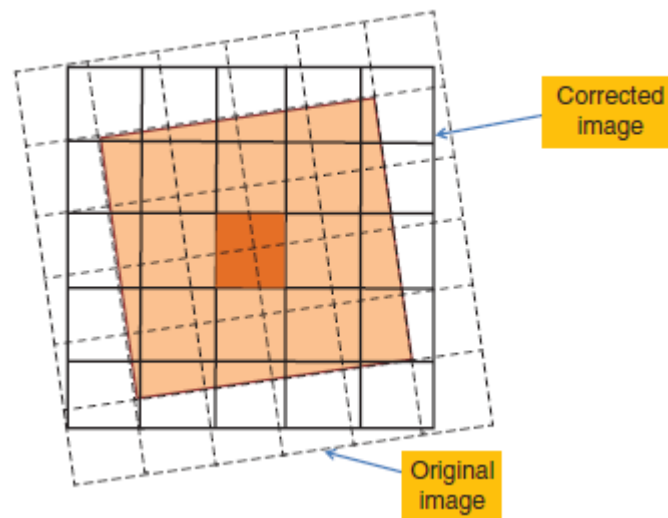


Figure 8.4 The cubic convolution method of image resampling.

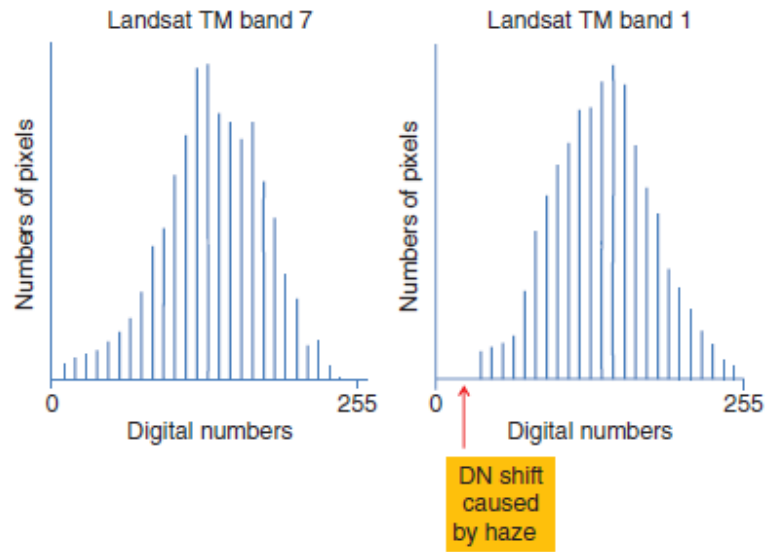


Errors or inconsistencies in image brightness values may occur in raw images due to variations in scene illumination, viewing geometry, atmospheric conditions, and sensor noise and response characteristics. These errors or inconsistencies, collectively known as “radiometric noise,” vary with the type of sensor and platform used to acquire the data and the conditions during data acquisition. Radiometric corrections aim at reducing the influence of the noise, and facilitating image interpretation and analysis as well as quantitative measurements from remotely sensed images. This section focuses on discussion of radiometric correction for electro-optical sensors because they are more frequently used.

For each data acquisition, the Earth-sun-sensor geometry relationship is different. As a result, variations in scene illumination and viewing geometry between images can be corrected by modeling the geometric relationship and distance between the area of the Earth’s surface imaged, the sun, and the sensor (Canada Centre for Remote Sensing, 2007). Image brightness magnitudes must be balanced or “normalized” across and between image scenes and spectral bands, and between different image acquisition dates. This procedure is especially needed to create an image mosaic of a large area that requires multiple image scenes to cover or to study changes in the surface reflectance over time. For such applications, corrections for sun elevation and the Earth–sun distance become important. The sun elevation correction can be implemented by dividing each pixel value in an image by the sine of the solar elevation angle for the time and location of image acquisition. The correction normalizes for the seasonal position of the sun relative to the Earth, assuming the sun was at the zenith when remote sensing occurred (Lillesand et al. 2008). Further, the correction needs to be carried out to account for the seasonal change of the Earth–sun distance, because the irradiance from the sun reduces on the order of the square of their distance (Lillesand et al., 2008).

In **Chap. 2**, we discuss how the atmosphere affects electromagnetic energy through processes of absorption, scattering, and reflection. Either absorption or scattering reduces the energy illuminating the ground surface. This atmospheric attenuation can be modeled mathematically to compute the amount of energy reaching a ground target, given that the information about atmospheric conditions during the image acquisition is available. Atmospheric scattering introduces “haze” to an image, and thus reduces image contrast. Various methods have been developed for haze compensation. One method is to examine the observed brightness values in an area of shadow or for a very dark object (such as a large, clear lake). The reflectance from these features in the near-IR region of the spectrum is assumed to be zero. If we observe minimum digital number (DN) values much greater than zero in another band over such an area or object, they are considered to have resulted from atmospheric scattering. This offset value can be subtracted from all pixels in that band (Sabins, 1997; **Fig. 8.5**).

Figure 8.5 Histograms for Landsat TM bands 7 and 1. The lack of pixels with low DNs in band 1 is caused by illumination from light selectively scattered by the atmosphere. (After Sabins, 1997.)



If a single-date image is used for image classification, atmospheric correction may not be required (Song et al., 2001) because signals from the terrestrial features are much stronger than those from the atmosphere. However, if a project requires estimating biophysical parameters from the Earth's surface, such as biomass, sediment, and land surface temperature, subtle differences in the surface reflectance are significant. In that case, conversion of DNs to absolute radiance values becomes a necessary procedure. Each sensor has its own response characteristics and the way in which the analog signal is converted to a digital number (analog-to-digital) conversion. Conversion of DNs to known (absolute) radiation or reflectance units also facilitates comparison between image data, especially for comparative analysis of several images taken by different sensors (for example, TM sensor on Landsat-5 versus ETM+ sensor on Landsat-7).

When multi-temporal or multi-sensor data are used together for an image classification, correction for atmospheric effects is also necessary. In such applications, relative radiometric correction is often applied to normalize the intensities among different bands within an image and to normalize the intensities of bands of one date to a standard scene chosen by the analyst (Jensen, 2005). The correction can be done by single-image adjustment using histogram adjustment (Fig. 8.5), or by multiple-data image normalization using regression. The latter method relates the DNs of the subject scene to those of the reference image band by band to produce a linear regression equation, and the coefficients and intercept of the equation are then used to compute a normalized image (Jensen, 2005). It should be noted that only the DNs of normalization targets in both scenes are used to develop the regression equation. The normalization targets should be chosen so as not to change in reflectance over time (Eckhardt, Verdin, and Lyford, 1990).

Errors or irregularities that occur in the sensor response or data recording and transmission can cause noise in an image such as systematic striping or banding and dropped lines (Canada Centre for Remote Sensing, 2007). Striping is caused by the variations and drift in the response over time among the detectors, which was common in early Landsat MSS data. The "drift" can be different among the six MSS detectors, causing the same brightness to be represented differently by each detector and thus an overall "striped" appearance (Canada Centre for Remote Sensing, 2007). Dropped lines occur when systematic errors result in missing or defective data along a scan line. The relative radiometric correction procedure is frequently applied to remove detector response error, line dropouts, striping, or line-start problems. On May 31, 2003, the ETM+ Scan Line Corrector (SLC) failed permanently, resulting in approximately 22% of the pixels per scene not being scanned. Although it is still capable of acquiring useful image data with the SLC turned off, particularly within the central part of any given scene, the scientific communities have developed a number of methods to fill in the data gaps (USGS, 2004; Roy et al., 2008; Pringle, Schmidt, and Muir, 2009).

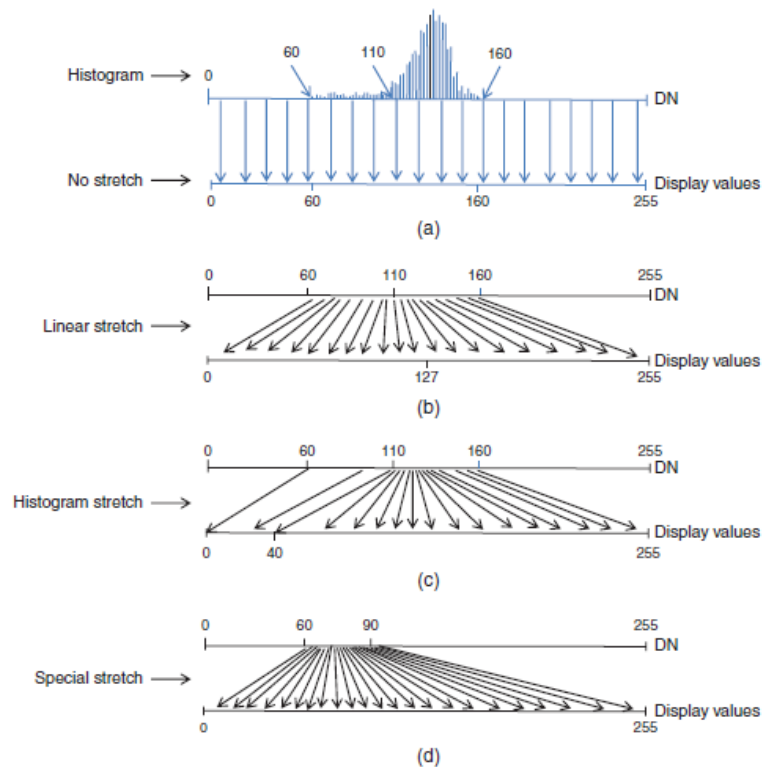
8.3. Contrast Enhancement

Various image enhancement methods may be applied to enhance visual interpretability of remotely sensed data as well as to facilitate subsequent thematic information extraction. Image enhancement methods can be grouped roughly into three categories: contrast enhancement, spatial enhancement, and spectral transformation. Contrast enhancement involves changing the original values so that more of the available range of digital values is used for display and the contrast between targets and their backgrounds is increased (Jensen, 2005).

Remote sensors, especially satellite sensors, are typically designed to detect and measure all levels of reflected and emitted energy emanating from the Earth's surface materials, objects, and features. A diverse range of targets (e.g., forest, deserts, snowfields, water, asphalt surface, etc.) can produce large variations in spectral response. No target can utilize all brightness ranges of a sensor. When an image is displayed, it is common that the range of brightness values does not match with the capability of a color display monitor. Thus, a custom adjustment of the range and distribution of brightness values is usually necessary. The range of brightness values present on an image is referred to as contrast. Contrast enhancement is a method of image manipulation that makes the interested image features stand out more clearly by making optimal use of the colors available on the display or output device (Jensen, 2005).

A variety of methods has been developed for contrast enhancement. Three of the methods—linear contrast stretch, histogram equalization, and special stretch—are described next. Before the detailed discussion, we have to know an important concept called “histogram.” An image histogram is a graphical representation of the brightness values that comprise an image, with the brightness values displayed along the x-axis of the graph and the frequency of occurrence of each of the brightness values on the y-axis (Fig. 8.6a). By applying different methods of contrast enhancement, the image histogram will change accordingly. Linear contrast stretch linearly expands the original brightness values of the image to the full range of sensitivity of the display device. In our example (Fig. 8.6b), the original minimum brightness value in the histogram is 60, and the maximum brightness value is 160. These 100 levels occupy less than one-half of the full 256 levels available for display. The values of 0 to 59 and 161 to 255 are not displayed. A linear stretch uniformly expands this small range to cover the full range of values from 0 to 255 by matching the old minimum value (60) to the new minimum value (0) and the old maximum value (160) to the new maximum value (255) and by scaling proportionately the old intermediate values between the new minimum and maximum values. This enhancement will make light-toned areas in the image appear lighter and dark-toned areas appear darker, so that visual interpretation is easier and the subtle differences in spectral response are more obvious.

Figure 8.6 Commonly used methods of image contrast enhancement. (After Lillesand et al., 2008.)



Linear contrast stretch does not work well if we want to see more details in the brightness values from 60 to 160 because there are more pixels in this range. A method called histogram equalization stretch can be more effective. This stretch assigns more display values to the more frequently occurring portions of the histogram and less display values to the lower frequency of brightness values. All pixel values of the image are redistributed so that approximately an equal number of pixels will occupy each of the user-specified output ranges (Jensen, 2005). As seen in Fig. 8.6c, the details in the high frequently occurring brightness values are better enhanced relative to those areas of the original histogram where values occur less frequently. This enhancement method will reduce the contrast in very light or dark parts of the image associated with the tails of a normally distributed histogram (Jensen, 2005).

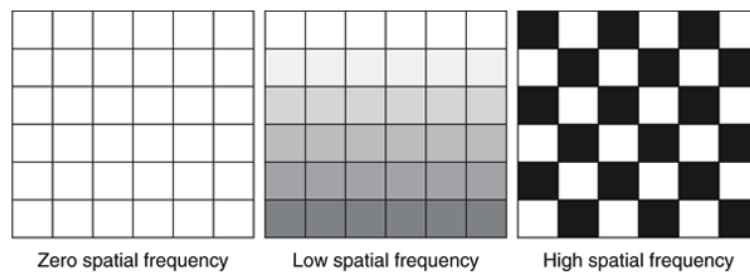
For a specific application, it may be desirable to assign the entire display range to a particular range of the image values. For example, vegetation features occupy only the digital values from 60 to 90, a very narrow range of the image histogram. We can enhance the characteristics of vegetation features by stretching that narrow range to the full display range from 0 to 255 (Fig. 8.6d). All pixels below or above these values would be assigned either to 0 or 255, and the details in these areas would be hidden. However, the brighter and greater contrast in vegetation features would be useful for delineation and analysis of vegetation.

8.4. Spatial Enhancement

While contrast enhancement focuses on optimal use of the colors available on the display or output device to match with image brightness values collected by a sensor, spatial enhancement is a group of image enhancement techniques that emphasize the spatial relationship of image pixels. These techniques explore the distribution of pixels of varying brightness over an image by defining homogeneous regions and by detecting and sharpening boundary discontinuities between adjacent pixels with notable differences in brightness values. The resultant images are often quite distinctive in appearance, and thus facilitate human recognition.

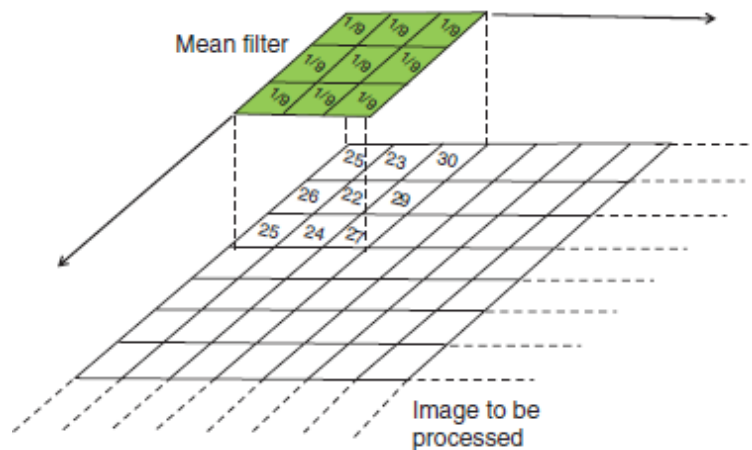
Various algorithms of spatial enhancement, such as spatial filtering, edge enhancement, and Fourier analysis, use the concept of *spatial frequency* within an image. Spatial frequency is the manner in which brightness values change relative to their neighbors in an image, which relates to the concept of texture that was discussed in [Chap. 4](#). Image areas of high spatial frequency, where the changes in brightness are abrupt over a small number of pixels, are of “rough” texture ([Fig. 8.7](#)). In contrast, “smooth” areas with little variation in brightness values over a relative large number of pixels have low spatial frequency. Many natural and man-made features in images have low spatial frequency, such as water bodies and large agriculture fields, while others exhibit high spatial frequency, such as geologic faults, edges of lakes, roads, and airports. Spatial filtering is the method that uses user-defined filters to emphasize or suppress image data of different spatial frequencies. Low-pass filters are designed to emphasize larger, homogeneous areas of similar brightness values and suppress local details in an image. In other words, low-pass filters are used to smooth (or to blur) the appearance of an image. High-pass filters, on the other hand, are designed to sharpen the appearance of local details in an image. A high-pass filter may be generated by subtracting a low-pass filter from the original image, leaving behind only the high spatial frequency information.

Figure 8.7 Examples of image spatial frequencies.



Spatial filtering involves passing a window of a few pixels in dimension (e.g., 3 pixels $\times$ 3 pixels) over each pixel in the image, applying a mathematical operation to the pixel values within that moving window, and creating a new image where the value of the central pixel in the window is the result of the mathematical operation ([Fig. 8.8](#)). A simple example of the moving window is the mean filter, with which a new image will have its pixel value as the average value of the closest nine pixels in the original image. In fact, spatial filtering belongs to a group of generic image processing operations called convolution, which has found itself numerous applications in remote sensing image analysis. The window (also known as kernel), a convolution operator, may be considered as a matrix (or mask) of coefficients that are to be multiplied by image pixel values to derive a new pixel value for a resultant enhanced image. The matrix may be of any size in pixels (e.g., 3 $\times$ 3, 5 $\times$ 5, 7 $\times$ 7) and different shapes (e.g., square, octagonal, nearest neighbor cross; Jensen, 2005). By changing the mathematical operations and the weightings (i.e., coefficients) of the individual pixels in the moving window, a variety of filters can be created to enhance or suppress specific types of features in the image. Cubic convolution, a resampling method as introduced in Sec. 8.2, is an example of the convolution technique. A more complex example involves making the kernel coefficients agree with a statistical distribution, such as a Gaussian distribution (Rajasekar and Weng, 2009).

Figure 8.8 Spatial filtering using a moving window.



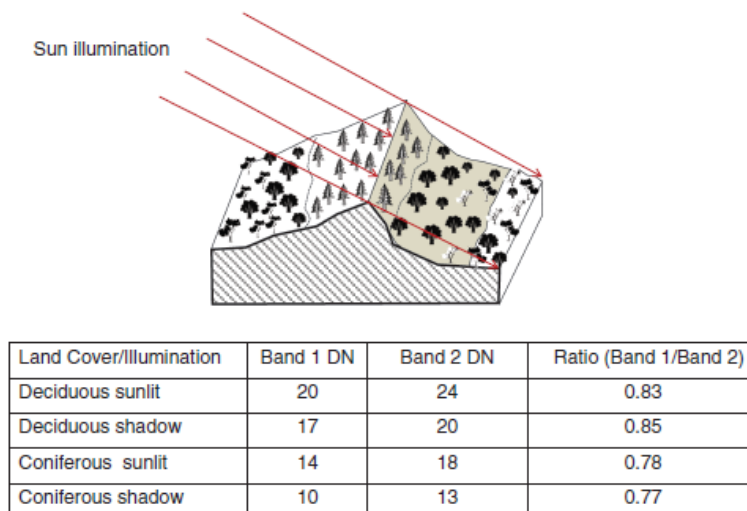
If we add a high-pass filter to the original image, a high-frequency enhancement will be achieved and an edge-enhanced image will be produced. Linear features (roads, geologic faults, etc.), composed of edges, are often not immediately recognizable from an image. Edge enhancement techniques are designed to highlight linear features, by expanding the width of linear features and by increasing the brightness difference across the features (Sabins, 1997). Digital filters, designed to enhance features oriented toward specific directions, are known as directional filters. These filters are especially useful in geological studies such as detecting linear geologic structures, and are frequently applied to radar images for increasing the visibility of linear features or lineaments (Arnold, 2004). Directional first differencing is one of the directional filtering techniques (Avery and Berlin, 1992; Lillesand et al., 2008). The other category of edge-enhancement filters is non-directional. To this category belongs the Laplacian filter, the most popular non-directional filter. They enhance linear features in all directions, while retaining the original image quality. The only exception is that those linear features oriented parallel with the direction of filter movement are not enhanced (Sabins, 1997).

Any remote sensing image may be considered to consist of several dominant spatial frequencies. We can use a mathematical technique called *Fourier Transform* to separate an image into its various spatial frequency components. If the pixel brightness values (DNs) in an image are plotted along each row and column, we will see numerous "peaks and valleys," corresponding to the high and low DN values. Therefore, an entire image looks like a complex wave, but mathematically, it can be broken down into simple sine and cosine waves of varying frequency, amplitude, and phase. We may then select and modify its components to emphasize certain groups of frequencies relative to others, and re-combine the image components into an enhanced image at the end. The modification is done through frequency convolution (not the same as the kernel convolution), which is a simple multiplication of an image mask designed by an image analyst, multiplied by a component image. Fourier transform has wide applications in noise removal from an image and image restoration.

8.5. Spectral Transformations

Spectral transformation refers to the manipulation of multiple bands of data to generate more useful information, and involves such methods as band ratioing, vegetation indices, principal component analysis, and so on. Band ratioing is often used to highlight subtle variations in the spectral responses of various surface covers, which may be masked by the pixel brightness variations from individual bands or in standard color composites. Ratioing is accomplished by dividing the DN value of one band by the value of another band. This quotient yields a new set of numbers that may range from 0 to 255, but the majority of the quotients will be fractional (decimal) values between 0 and typically 2 to 3 (Short, 2009). It is important to stretch the resultant ratio image for a better visualization, based on the computer display limits. It is common to multiply the quotients by a normalization factor of 162.3 for an 8-bit display (Lillesand et al., 2008). Brighter tones in the ratio image indicate larger differences in spectral response between the two bands. An effect of band ratioing is that variations in scene illumination because of topographic effects are minimized. The image brightness values are lower in the shadowed side of a hill, but are higher in the sunlit side (Fig. 8.9). However, the ratio values for a specific feature will always be highly similar regardless of the illumination conditions. It is important to remove image noise and atmospheric haze before implementing image ratioing. This is because that the ratioing process cannot remove these additive factors. With a multispectral dataset, we can produce various forms and a number of ratio combinations. For example, with a Landsat MSS dataset of four spectral bands, a total of six ratioed images (band 4/5, 4/6, 4/7, 5/6, 5/7, 6/7) and six reciprocals (band 5/4, 6/4, 7/4, 6/5, 7/5, 7/6) can be generated. In addition to inter-band ratios, an individual band can also be divided by the mean value of all bands in an image dataset, which will result in a normalized band of the particular band (Sabins, 1997). The utilization of any ratio has to consider the particular reflectance characteristics of the features involved and the application of a specific project (Lillesand et al., 2008).

Figure 8.9 An illustration of the effect of band ratioing. Variations in scene illumination because of topographic effects are minimized. (After Lillesand et al., 2008.)



Various forms of ratio combinations have been developed as vegetation indices for assessing and monitoring green vegetation conditions. Healthy green vegetation reflects strongly in the near-IR portion of the electromagnetic spectrum, while absorbing strongly in the visible red spectrum. This strong contrast provides the basis for developing quantitative indices of vegetation conditions. A vegetation ratio image, for example, Landsat TM band 4 (near-IR infrared—0.76 to 0.90 μm) divided by band 3 (red—0.63 to 0.69 μm), would result in values much greater than 1.0 for healthy vegetation, but a value close to 1.0 for soil and water. Soil and water yield similar brightness values in both the near-IR and red bands. We can also use vegetation indices to discriminate between stressed and non-stressed vegetation. Vegetation under stress will have a low near-IR reflectance, and therefore the ratio values will be smaller than healthy green vegetation. The most popular form of vegetation index is the normalized difference vegetation index (NDVI), which is produced by dividing the difference between the two bands (near-IR and red) by their sum. The NDVI value ranges from -1 to 1 , with 0 representing the approximate value of no vegetation and negative values representing non-vegetated surfaces. NDVI has been used in a variety of contexts to assess green biomass and as a proxy to overall health of ecosystems. In [Chap. 7](#), we illustrate two applications of NDVI, one for monitoring global vegetation conditions using a MODIS sensor ([Fig. 7.10](#)) and the other for monitoring vegetation conditions in cropland and pasture regions throughout the growing season in the United States by using NOAA-18 NDVI weekly composites ([Fig. 7.15](#)).

Different bands of a multispectral data set often contain similar information. If you compare individual TM bands of the Terre Haute image ([Fig. 3.12](#) in [Chap. 3](#)), you will note that band 1 and band 2 images look similar, as do bands 5 and 7 images. However, band 4 is unique, showing strong contrast with other bands, especially with band 1. These visual and numerical similarities in brightness indicate that some bands are highly correlated. Therefore, there are some redundancies in a multispectral dataset. A statistical technique, known as principal component analysis (PCA), can be used to reduce the data redundancy so that more efficient data analysis and easier interpretation is possible. To explain this mathematical transformation, we can plot pixel values from two bands on a scatter diagram ([Fig. 8.10](#)). The transformation defines a new axis (Axis 1) oriented in parallel with the long dimension of the scatter and the second axis perpendicular to Axis 1. These new axes define the directions of principal components, and the principal component image data values are computed as a linear combination of pixel values in the original coordinate system. The same transformation may be applied to a multispectral dataset of any number of bands, or a hyperspectral image. The process typically results in fewer principal components uncorrelated with one another, which are ordered in terms of their explanatory power (higher-order components explain more information; [Fig. 8.11](#)). Noise is often denoted as a less-correlated component image (Sabins, 1997). [Figure 8.12](#) illustrates the result of principal component analysis derived from a hyperspectral image. The first component resembles a black-and-white aerial photograph, revealing the information on albedo (the ratio of reflected radiation to incident radiation upon a surface). The second component is mainly associated with healthy vegetation, while the third component is associated with bare soils and grounds. Principal components are usually easier to interpret visually than the original bands. Digital analysis of these component images are also found more efficient and effective. PCA has been used frequently either as a spectral enhancement technique to enhance visual interpretation or as a feature extraction technique for subsequent use in digital classification procedures.

Figure 8.10 Rotation of axes in the principal component transformation (example of 2D data).

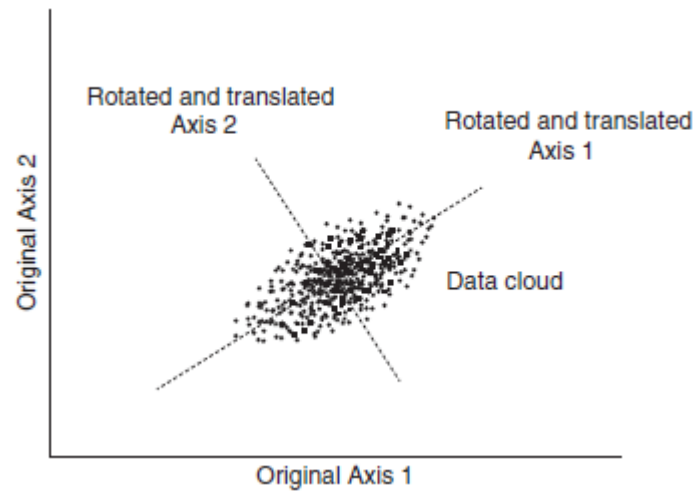


Figure 8.11 Illustration of principal component analysis for a Landsat TM image.

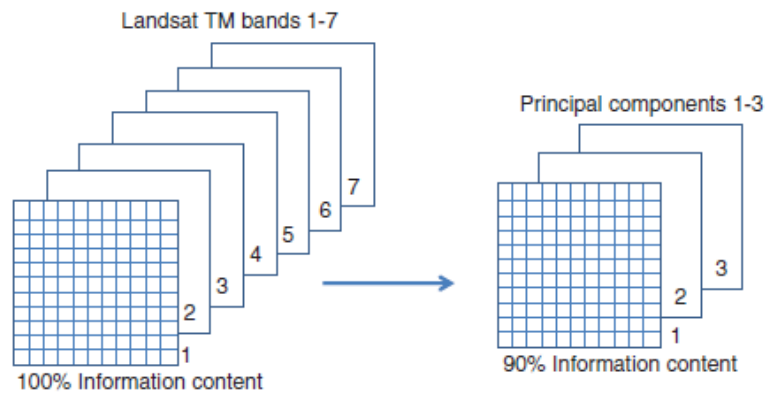
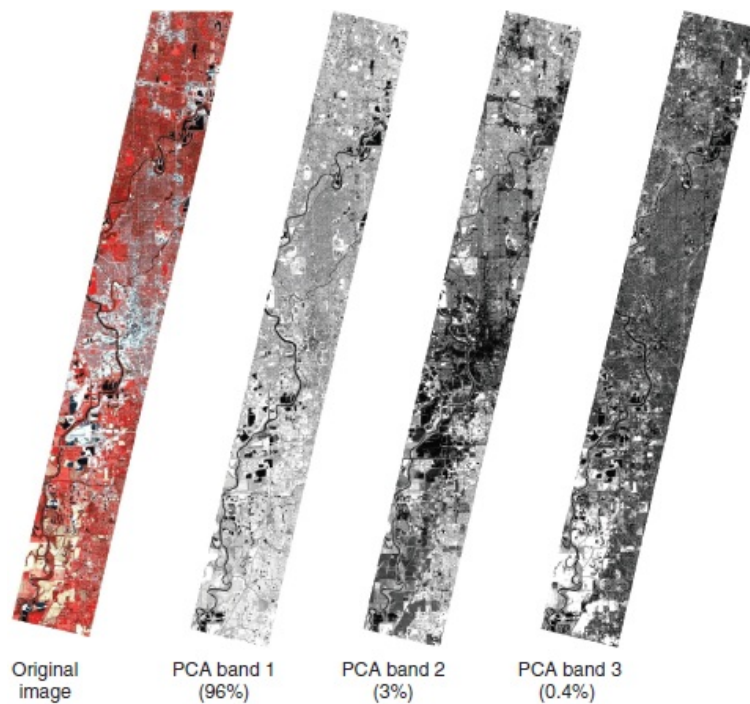


Figure 8.12 An illustrated result of principal component analysis for a hyperspectral image.



8.6. Image Classification

8.6.1. Image Classification Process

Different features usually have different spectral response characteristics, which produces a unique combination of digital numbers in various bands for each pixel. Image classification uses spectral information represented by the DNs in one or more spectral bands and attempts to classify each individual pixel based on the spectral information. The objective is to assign all pixels in the image to particular classes or themes (e.g., water, vegetation, residential, commercial, etc.) and to generate a thematic “map.” Depending on the goal of a specific project, an information class may be divided further into subclasses. For example, vegetation can be broken down into forest, cropland, grassland, pasture, and so on. It is important to differentiate between information classes and spectral classes. The former refers to the categories of interest that the analyst is actually trying to identify from the imagery, while the latter are the groups of pixels that are uniform (or near similar) with respect to their brightness values in the different spectral channels of the data. The ultimate goal of image classification is to derive information classes of interest. Although some digital image processing techniques may be applicable to both multispectral and hyperspectral imagery, we focus our discussion on multispectral imagery in this section and save hyperspectral image analysis for the next section.

Feature extraction is sometimes necessary for subsequent image classification. It serves as a pre-processing step to reduce the spectral, or spatial, dimensionality of an image. Feature extraction can be accomplished by using any type of spectral or spatial transforms to reduce the data or to enhance its multispectral features, or even by simply selecting a subset of multispectral bands. Many potential variables may be used in image classification, including spectral signatures, vegetation indices, transformed images, textural or contextual information, multi-temporal images, multi-sensor images, and ancillary data. Because of different capabilities in spectral separability, the use of too many variables in a classification procedure may decrease classification accuracy (Price, Guo, and Stiles, 2002). It is important to select only the variables that are most effective for separating thematic classes. Selection of a suitable feature extraction approach is especially necessary when hyperspectral data are used. This is because of the huge amount of data volume and high correlations that exist among the bands of hyperspectral imagery, and also because a large number of training samples is required in image classification. Many feature extraction approaches have been developed, including principal component analysis, minimum noise fraction transform, discriminant analysis, decision boundary feature extraction, non-parametric weighted feature extraction, wavelet transform, and spectral mixture analysis (Lu and Weng, 2007).

Before starting an image classification procedure, it is often helpful to generate color composites of various combinations of bands. A spacecraft (Landsat, SPOT, etc.) collects multiple bands of images in black and white, not in color. An interpreter can only be able to distinguish between 10 and 20 shades of grey; however, he or she may be able to distinguish many more colors. Color composites help to identify various features and make interpretation easier. In displaying a color composite image on the screen, computers use the light of three primary colors (red, green, and blue), just like color TVs. The three-color lights can be combined in various proportions in order to produce various colors (Fig. 8.13). A color composite image is generated by assigning each primary color to a separate spectral band (Fig. 8.14). Depending on how well the colors in the image composites match with colors we see in nature, we may produce false and true/natural color composites. A common form of false color composite is generated by projecting a green band image through the blue color gun, a red band through the green color gun, and a near-IR image through the red color gun (Fig. 8.14). A false color composite is valuable to interpret vegetation. Vegetation appears in different shades of red due to its high reflectance in the near-IR band, and the shades vary with the types and conditions of the vegetation (Liew, 2001). In a true or natural color composite image, colors resemble a visible color photograph, that is, vegetation in green, water in blue, soil in brown, and so on (Fig. 8.15). A true/natural color image can be generated by projecting blue, green, and red spectral bands through the blue, green, and red color guns, respectively. Many other forms of false and natural color composites via other combinations of bands and color lights may bring out or call attention to individual scene features that are usually present in more subtle expressions (Short, 2009). Furthermore, ratio images can also be used to generate color composites by combining three black-and-white ratio images. This type of color composite favors incorporating more than three bands and facilitates the interpretation of subtle spectral differences (Lillesand et al., 2008). Similarly, it is common to form a color composite using three of the resultant principal component images.

Figure 8.13 Three primary colors (red, green, and blue) and other colors by combining different intensities of these three colors.

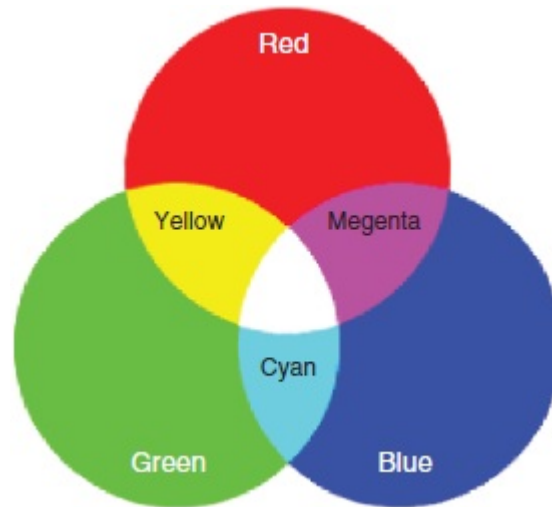


Figure 8.14 Process for creating a false color composite using Landsat TM bands.

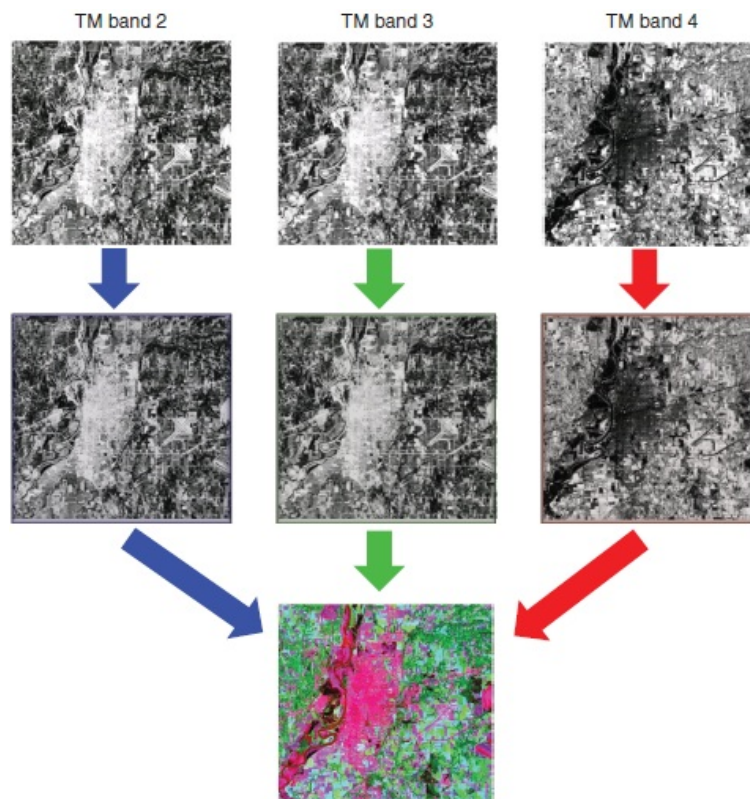
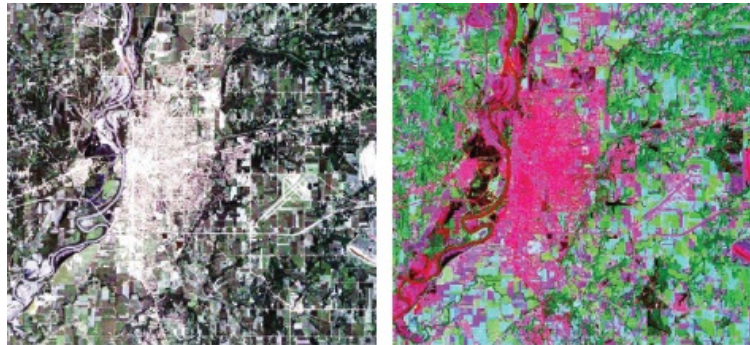


Figure 8.15 A comparison of a true/natural color with false color composite.



Generally, there are two approaches to image classification: supervised and unsupervised classification. In a supervised classification (Fig. 8.16), the analyst identifies in the imagery homogeneous representative samples of different cover types (information classes) of interest to be used as training areas. Therefore, the image analyst should be familiar with the geographical area and the actual surface cover types present in the image by experience with thematic maps or by on-site visits. In this sense, the analyst is “supervising” the classification of a set of land cover types. For each thematic class, multiple training areas can be selected. The spectral signature will then be determined for the class based on the DNs for each band from all the pixels enclosed in the train areas. Once numerical spectral signatures have been developed for all the thematic classes, each pixel in the imagery would then compare spectrally with these signatures to determine to which information class it should belong. Various algorithms of supervised classification may be used to assign an unknown pixel to one of a number of the thematic classes, among which are the minimum distance, parallelepiped, and maximum likelihood. In an unsupervised classification, spectral classes are first grouped based solely on digital numbers of the band selected for use (Fig. 8.17). Various statistical methods can be used to cluster all pixels in an image into spectral groups. The analyst needs to specify the number of groups to be looked for in the image. Later, these spectral groups can be combined, disregarded, or broken down into smaller groups. Because the identities of the spectral groups are not initially known, the analyst requires matching them with information classes by using some form of references (e.g., a thematic map or large-scale aerial photographs).

Figure 8.16 The process of a supervised image classification.

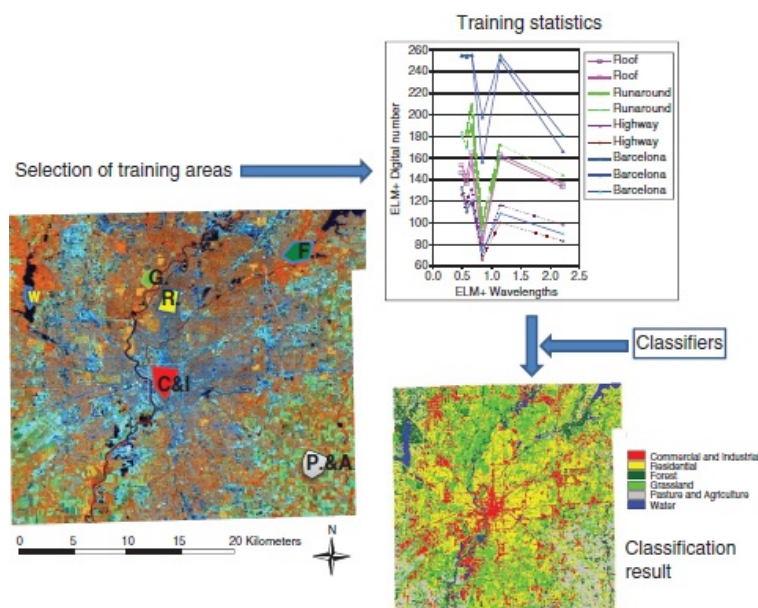
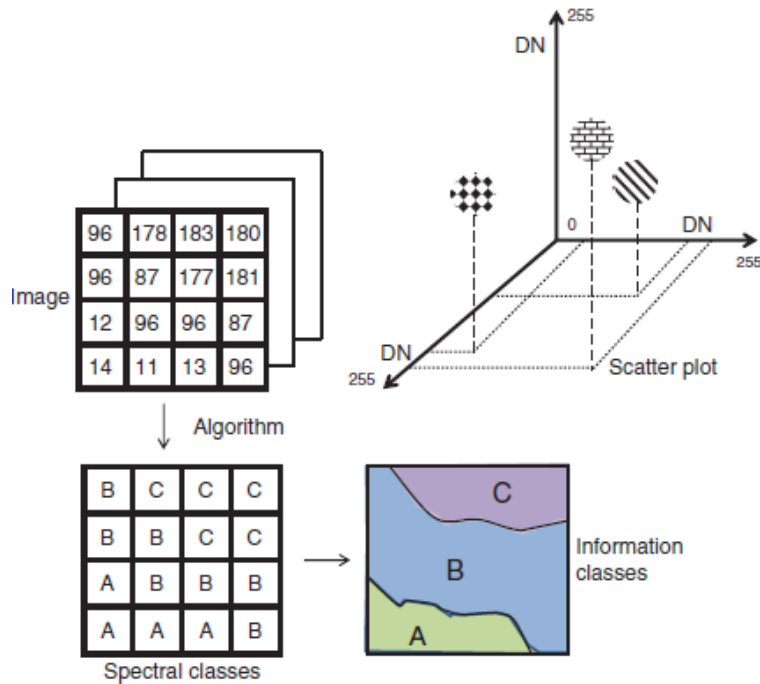


Figure 8.17 An illustration of the process of an unsupervised image classification.

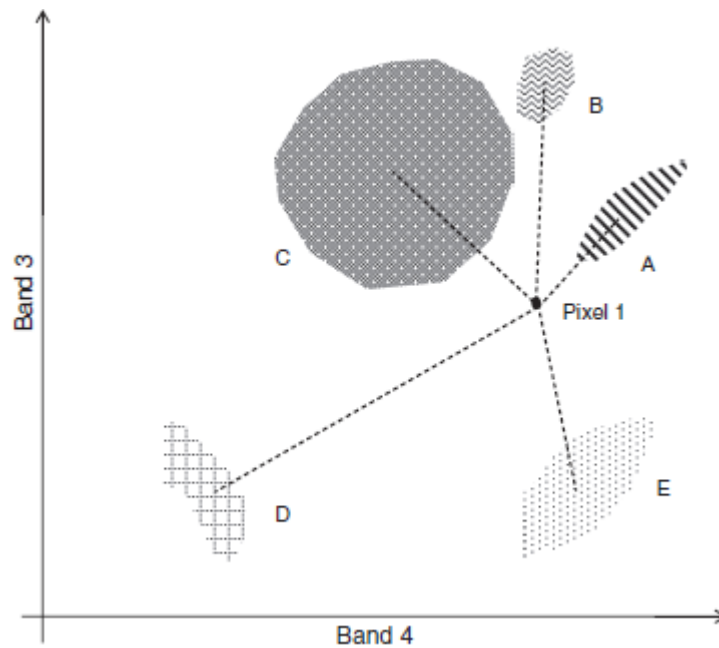


8.6.2. Common Image Classifiers

An image classifier is a computer program that implements a classification procedure (Campbell, 2007). Various classifiers of supervised classification have been developed for assigning an unknown pixel to one of the information classes. Among the most frequently used are the minimum distance, parallelepiped, and maximum likelihood classifiers.

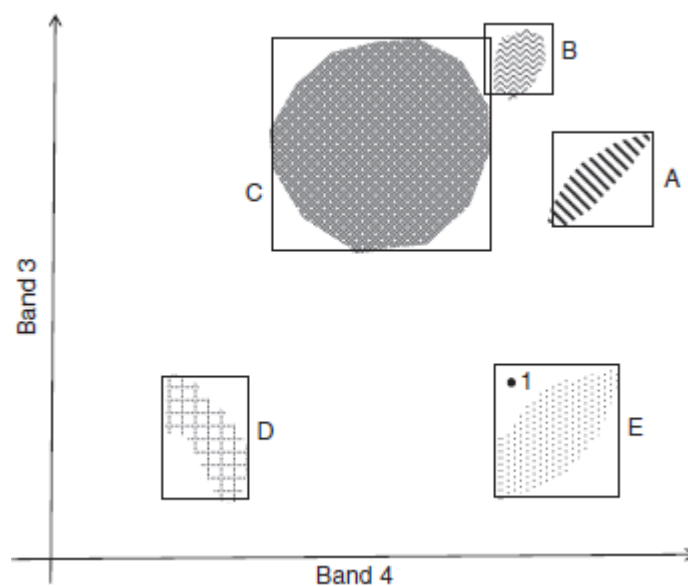
The minimum distance classifier computes the mean vectors of all bands for each class based on the training data sets. An unknown pixel is classified by computing the distance between the DN value of this unknown pixel and each mean vector. The unknown pixel is then assigned to a class to which it has the minimum distance in the spectral space. In Fig. 8.18, pixel 1 is assigned to class A. An analyst may specify a threshold value of distance for each class. If a pixel shows a distance from the mean vectors of all classes farther than the specified thresholds, it will be classified as "unknown." The minimum distance classifier is computationally simple, but it is not sensitive to different degrees of variance in the spectral response data. Therefore, the classifier is not applicable where spectral classes are close in the feature space and are of high variance.

Figure 8.18 An illustration of the minimum-distance-to-means classifier (Pixel 1 is categorized to belong to class A).



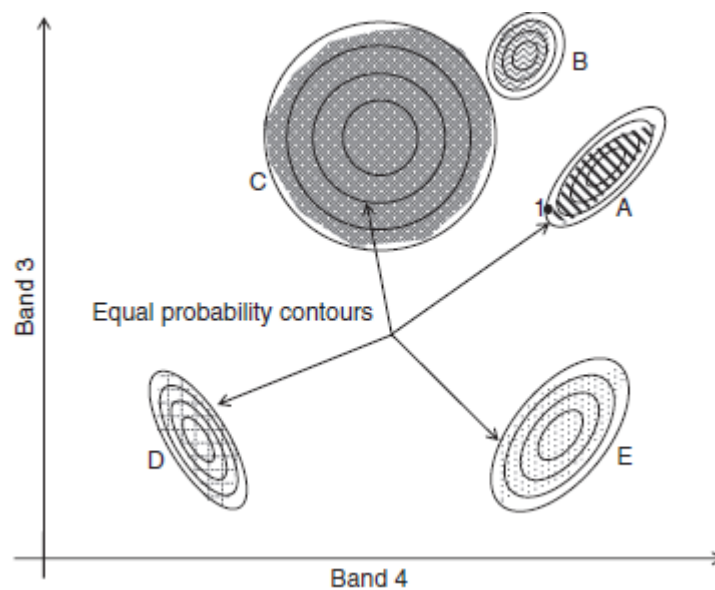
The parallelepiped classifier first calculates the range of DN values in each category of training data sets. An unknown pixel is assigned to a class in which it lies. If a pixel is located outside all class ranges, it will be classified as “unknown.” In Fig. 8.19, pixel 1 will be assigned to class E. The parallelepiped classifier also has the advantage of being computationally simple. Its main difficulty arises when two or more category ranges overlap. Unknown pixels that lie in the overlapped areas will be assigned arbitrarily to one of the overlapping classes or be labeled as “unknown.” The overlapping in the feature space often indicates that two classes correlate spectrally or have high covariance, tendency of spectral values to vary similarly in two bands (Lillesand et al. 2008). It may also be caused by bad selection of training sets for one or more classes.

Figure 8.19 An illustration of the parallelepiped classifier (Object 1 is categorized as class E). Note: Class B and class C are spectrally overlapped.



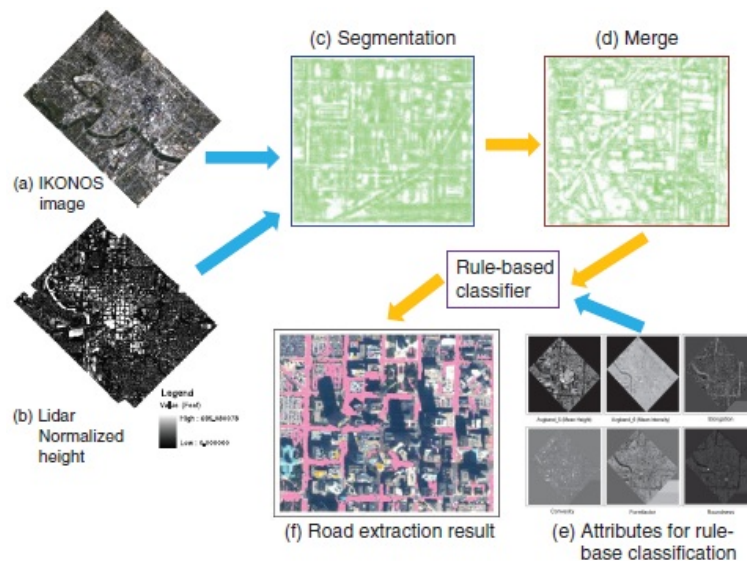
The maximum likelihood classifier assumes that the training data statistics for each class in each band are normally distributed, that is, the Gaussian distribution. Under this assumption, the distribution of spectral response pattern of a class can be measured by the mean vector and the covariance matrix. A probability density function value is calculated for each class according to these parameters. On a 2D scatter plot, the class DN distribution gives rise to elliptical boundaries, which define the equal probability envelope for each class. **Figure 8.20** shows equal probability contours for each class. To classify an unknown pixel, the maximum likelihood classifier computes the probability value of the pixel belonging to each class, and assigns it to the class that has the largest (or maximum) value. A pixel will be labeled “unknown” if the probability values are all below the threshold values set by the analyst. This classifier is the most accurate one among the three classifiers we introduce here because it evaluates both the variance and the covariance of the category spectral response patterns. However, if the assumption of data normality is not met for any class, the accuracy of classification result will suffer.

Figure 8.20 An illustration of the maximum likelihood classifier (Object 1 falls within the contour of class A, and is categorized as class A).



Many advanced classification approaches, such as artificial neural network, fuzzy-set, and expert systems, have been developed for image classification. A description of each category is beyond the scope of this discussion. Readers who wish to have a better knowledge of them may like to look up the article by Lu and Weng (2007). One classification method—object-oriented classification—has recently become quite popular (Gitas, Mitri, and Ventura, 2004; Walter, 2004). Two steps are involved in the classification process (**Fig. 8.21**). Image segmentation merges pixels into objects, and a classification algorithm is then applied to the objects instead of to the individual pixels. In the process of creating objects, the analyst must control the segmentation scale, which determines the occurrence or absence of an object class, and the sizes of objects. Both parameters affect a classification result. This approach has proven to be able to provide better classification results than per-pixel-based classification approaches, especially for fine spatial resolution data (mostly better than 5 m spatial resolution, such as IKONOS and QuickBird images).

Figure 8.21 The process of an object-based image classification for road extraction by using the combined datasets of IKONOS image and Lidar data. The road extraction process includes mainly three steps: segmentation, merging, and rule-based extraction.

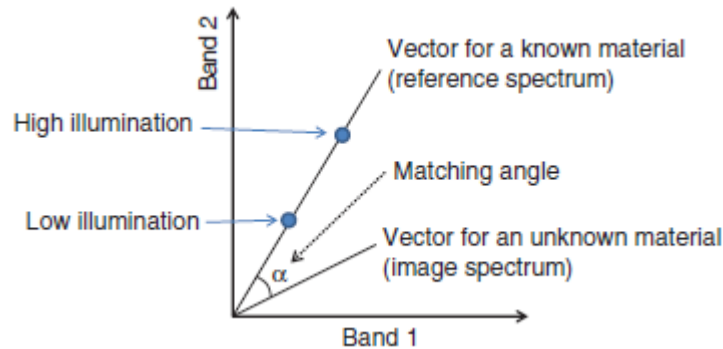


8.7. Hyperspectral Image Analysis

In [Chap. 3](#), we discussed the main difference between a multispectral and a hyperspectral imaging system. While multispectral sensors typically provide discrete measurements of average reflected or emitted radiance over isolated, wide spectral bands, hyperspectral sensors produce contiguous, high spectral-resolution radiance spectra. Therefore, multispectral images can only discriminate general differences among surface material types, but a hyperspectral sensor has the potential for detailed identification of materials and better estimates of their abundance. Many techniques for hyperspectral image analysis are derived from the field of spectroscopy, which was developed for mineral mapping in the early 1980s (Goetz et al., 1985). The molecular composition of a particular material results in the distinctive patterns in which the material absorbs and reflects energy at individual spectral wavelengths (Lillesand et al., 2008). Many minerals have unique spectral reflectance signatures, as do soils. Plants display diagnostic “peaks” and “valleys” along their spectral reflectance curves because of the presence of pigments, water, and other chemical constituents. The large number of very narrowly sampled hyperspectral bands allows the use of the remote sensing data to compare with and to replace the expensive measurements obtained in laboratories or from fields. These measurements are often stored in the digital “libraries” of spectral reference data, primarily for minerals, soils, vegetation types, and man-made materials, and are used as reference data in hyperspectral image analysis.

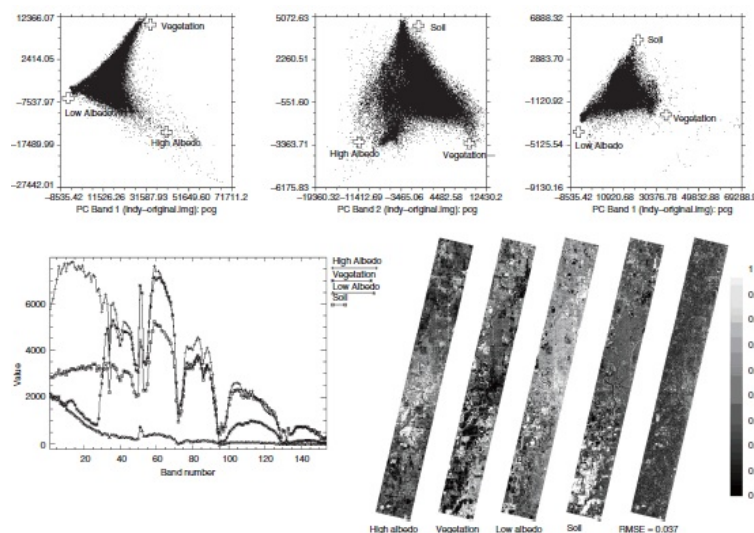
Various techniques have been developed for spectral matching between image spectra and the reference spectra. The spectral angle mapper (SAM; Kruse, Lefkoff, and Dietz, 1993) is a commonly used method for comparing an image spectrum to a reference spectrum. This method treats both spectra as vectors in a multidimensional space, and the number of dimensions equals the number of spectral bands ([Fig. 8.22](#)). To compare the two spectra, the spectral angle between them is computed. If the angle is smaller than a set threshold, the image spectrum is considered to match with the reference spectrum. This method does not consider the illumination condition, which modulates the vector length. The result of the SAM is an image showing the best match at each pixel.

Figure 8.22 An illustration of spectral angle mapping concept.



Spectral mixture analysis (also known as spectral unmixing) is another technique widely applied in analyzing hyperspectral remote sensing images. Most Earth surfaces are composed of various materials, resulting in mixing of spectral signatures within a single pixel. Depending on how the materials are arranged on the surface, the mixing of the signatures can be represented as a linear model, or a more complex, non-linear model. Spectral mixture analysis assumes that mixed spectral signatures are caused by the mixture of a set of “pure” materials, called end members (Fig. 8.23), whose spectra can be measured in the laboratory, in the field, or from the image itself. Once the spectral mixing model is determined, it can be used to invert the proportions of those “pure materials.” The outputs of spectral unmixing are “abundance” images for each end member, which show the fractions of each pixel occupied by each end member. Because the maximum number of end members that can be included is directly proportional to the number of spectral bands plus one, the vastly increased dimensionality of a hyperspectral sensor allows deriving a great deal of “pure” materials, and selects only those bands with high SNRs or with less atmospheric absorption effects (Lillesand et al., 2008).

Figure 8.23 Scatterplots of the principal components showing the locations of potential endmembers. Spectral reflectance characteristics of the selected endmembers are also shown. Fraction images (high albedo, low albedo, vegetation, and soil) and error images from spectral mixture analysis of the ALI and Hyperion images.



Spectral transform techniques, such as ratioing, vegetation indices, and principal component analysis, have been widely applied in hyperspectral image analysis. Because of the huge number of spectral bands, we have great flexibility to choose spectral bands for use in ratioing. With multispectral data, we have only one choice of using the red and near-IR bands for generating vegetation ratio, while with hyperspectral data, we can have many choices for producing such red and near-IR narrow band combinations (Gong, Pu, Biging, and Larrieu, 2003). Conventional multispectral classification techniques, such as maximum likelihood classifier, cannot be applied to hyperspectral image data due to the high dimensionality of the data. These classifiers require that every training set must include at least one more pixel than there are bands in the sensor, and that it is generally desirable to have the number of pixels per training set be between 10 and 100 times as large as the number of sensor bands to derive statistically meaningful training sets (Benediktsson, Sveinsson, and Arnason, 1995; Lillesand et al., 2008). To overcome these problems, feature extraction becomes necessary before image classification. Feature extraction techniques, such as principal component analysis, or its noise-adjusted version, maximum noise fraction, have been applied in transforming and reducing the data dimension by maximizing the ordered variance of the whole data set or the ratio of between-class variance and within-class variance of the training samples (Pu and Gong, 2011). The maximum noise fraction transform maximizes the SNR in choosing principal components (Green, Berman, Switzer, and Craig, 1988). Principal components resulted from these transforms can be used in subsequent image classification and for determining end member spectra for spectral mixture analysis. Alternatively, a non-parametric classifier, such as a neural network, can be used for classification of hyperspectral images (Filippi and Jensen, 2006).

8.8. Summary

Digital image processing goes far beyond what we have discussed in this chapter. In fact, it is a common interest of multiple fields, such as remote sensing, pattern recognition, signal processing, and computer vision. From a remote-sensing perspective, digital image processing copes with subtle variations in brightness values (DNs). Thus, we are concerned about image data characteristics, especially radiometric, spatial, and spectral qualities. Since most techniques of image corrections and enhancements, as well as feature extraction techniques, do alter image brightness values, we should always select carefully before implementing any technique. For a long time, image classification mainly utilized brightness values of individual pixels in multi-spectral bands. These types of image classifications have been termed as “spectral classifiers” (Campbell, 2007) or “spectral pattern recognition” (Lillesand et al., 2008). As high spatial-resolution data become readily available today, image texture has been used in image classification, as well as contextual information, which describes the association of neighboring pixel values. In this chapter, we introduced a new classification method known as “object-based image analysis.” The stereoscopic viewing capacity of the high-resolution images, such as IKONOS and QuickBird, further stimulates the use of manual interpretation methods and photogrammetric techniques parallel with digital image processing methods (Jensen, 2005; Tong, Liu, and Weng, 2010). Similarly, since the mid-1980s, various hyperspectral sensors continue to emerge, which provide a large amount of hyperspectral image data for applications in different fields. The birth of a unique set of techniques for hyperspectral remote sensing, such as those discussed in this chapter, become necessary. All methods and techniques discussed in this chapter are suitable for both reflected and emitted bands of digital images. However, there are some particular methods suited only for the emitted bands, which will be discussed in [Chap. 9](#).

8.9. Key Concepts and Terms

Absolute radiometric correction Conversion of image DNs to absolute radiance values to account for a specific sensor’s response characteristics and the analog-to-digital conversion.

Band ratioing Dividing the DN value of one band by the value of another band.

Bilinear interpolation resampling Calculates the output cell value by calculating the weighted average of the four closest input cells based on distance.

Color composite image Image generated by assigning each primary color to a separate spectral band.

Contrast enhancement The histogram of DN values is modified so that more of the available range of digital values is used for display, and the contrast between targets and backgrounds is increased.

Cubic convolution resampling Calculates the output cell value by calculating the weighted average of the closest 16 input cells based on distance.

Digital image analysis The tasks for detecting, identifying, measuring, and analyzing features, objects, phenomena, and processes from remote sensing images using computers.

Directional filters Filters designed to enhance features oriented toward specific directions.

False color composite Commonly generated by projecting a green band image through the blue color gun, a red band through the green color gun, and a near-IR image through the red color gun. A false color composite is valuable to interpret vegetation.

Fourier transform A mathematical technique that separates an image into its various spatial frequency components. The components are modified through frequency convolution to emphasize certain groups of frequencies and recombine the image components into an enhanced image at the end. Fourier transform has wide applications in noise removal from an image and image restoration.

Geometric correction Image preprocessing that corrects systemic and nonsystematic errors contained in the remote sensing system and during image acquisition. It commonly involves digital rectification and resampling.

Ground control point (GCP) A specific pixel on an image or location on a map whose geographic coordinates are known.

High-pass filters Filters designed to sharpen the appearance of local details in an image.

Histogram equalization stretch This stretch assigns more display values to the more frequently occurring portions of the histogram and less display values to the lower frequency of brightness values. All pixel values of the image are redistributed so that approximately an equal number of pixels will occupy each of the user-specified output ranges.

Image classification Uses spectral information represented by the DNs in one or more spectral bands and attempts to assign all pixels in the image to particular classes or themes (e.g., water, vegetation, residential, commercial, etc.) and to generate a thematic "map."

Image histogram A graphical representation of the brightness values that comprise an image, with the brightness values displayed along the x-axis of the graph and the frequency of occurrence of each of the brightness values on the y-axis.

Linear contrast stretch Linearly expands the original brightness values of the image to the full range of sensitivity of the display device.

Low-pass filters Filters designed to emphasize larger, homogeneous areas of similar brightness values and suppress local details in an image. In other words, low-pass filters are used to smooth (or to blur) the appearance of an image.

Maximum likelihood classifier This algorithm assumes that the training data statistics for each class in each band are normally distributed, that is, the Gaussian distribution. Under this assumption, the mean vector and the covariance matrix can measure the distribution of spectral response pattern of a class. A probability density function value is calculated for each class according to these parameters. To classify an unknown pixel, the probability value of the pixel belonging to each class is calculated, and it is assigned to the class that has the largest (or maximum) value.

Minimum distance classifier Computes the mean vectors of all bands for each class based on the training data sets. An unknown pixel is classified by computing the distance between the DN value of this unknown pixel and each mean vector. The unknown pixel is then assigned to a class to which it has the minimum distance in the spectral space.

Nearest neighbor resampling Uses the input cell value closest to the output cell as the assigned value to the output cell.

Non-systematic errors of satellite images Errors caused by the sensor platform departing from its normal altitude or the change in terrain elevation, and by the failure of the axis of a sensor system to maintain normal to the Earth's surface and the other parallel to the platform's flight direction.

Object-oriented classification Two steps are involved in the classification process. Image segmentation merges pixels into objects, and a classification algorithm is then applied to the objects instead of individual pixels.

Parallelepiped classifier Calculates the range of DN values in each category of training data sets. An unknown pixel is assigned to a class in which it lies.

Principal component analysis A mathematical technique that transforms correlated variables (e.g., image bands) to fewer principal components uncorrelated to each other.

Radiometric correction Correction of errors or inconsistencies in image brightness value, that is, radiometric noise.

Radiometric noise Errors or inconsistencies in image brightness values occurring in raw images due to variations in scene illumination, viewing geometry, atmospheric conditions, or sensor noise and response characteristics. These errors or inconsistencies vary with the type of sensor and platform used to acquire the data and the conditions during data acquisition.

Rectification The process by which the geometry of an image is made planimetric.

Registration The process of geometrically aligning two or more sets of image data such that resolution cells for a single ground area can be digitally or visually superimposed.

Relative radiometric correction An image preprocessing procedure used to normalize the intensities among different bands within an image or to normalize the intensities of bands of one date to a standard scene chosen by the analyst. The correction can be done by single-image adjustment using histogram adjustment, or by multiple-data image normalization using regression.

Resampling The process of extrapolating data values to a new grid. It is the step in rectifying an image that calculates pixel values for the rectified grid from the original data grid.

Spatial enhancement A group of image enhancement techniques that emphasize the spatial relationship of image pixels. These techniques explore the distribution of pixels of varying brightness over an image by defining homogeneous regions and by detecting and sharpening boundary discontinuities between adjacent pixels with notable differences in brightness values.

Spatial filtering The method that uses user-defined filters to emphasize or to suppress image data of different spatial frequencies.

Spatial frequency The manner in which brightness values change relative to their neighbors in an image.

Spectral angle mapper A commonly used method for comparing an image spectrum to a reference spectrum that treats both spectra as vectors in a multidimensional space and the number of dimensions equals the number of spectral bands. The spectral angle between the two spectra is computed for determination of matching.

Spectral mixture analysis It assumes that mixed spectral signatures are caused by the mixture of a set of "pure" materials, called end members, whose spectra can be measured in the laboratory, in the field, or from the image itself. A spectral mixing model is developed to invert the proportions of those "pure materials."

Supervised classification The analyst identifies in the imagery homogeneous representative samples of different cover types of interest to be used as training areas. The spectral signature will then be determined for the class based on the DN's of training areas. Each pixel in the imagery would compare spectrally with these signatures to determine to which information class it should belong.

Systematic errors of satellite images Errors caused by scan skew, mirror-scan velocity variance, platform velocity, Earth rotation, and the curvature of the Earth's surface.

True or natural color composite A true/natural color image may be generated by projecting blue, green, and red spectral bands through the blue, green, and red color guns, respectively. Colors in this type of image composite resemble a visible color photograph, that is, vegetation in green, water in blue, soil in brown, and so on.

Unsupervised classification Spectral classes are first grouped based solely on digital numbers of the band using a statistical method. Later, these spectral groups can be combined, disregarded, or broken down into smaller groups. The identities of the spectral groups need to match with information classes by using some forms of references (e.g., a thematic map or aerial photographs).

8.10. Review Questions

1. Every satellite image is obtained in black-and-white at a precise wavelength. Why do some satellite images have different colors when seen on a computer screen?
2. Examine the Landsat TM color composite images of Terre Haute, Indiana, U.S.A. (Fig. 8.15). Identify one or two uses for each color composite.
3. Radiometric correction is not necessarily equally useful for all remote sensing applications. Can you illuminate on a situation where absolute radiometric correction should be conducted and another situation when relative radiometric correction is necessary?
4. For your particular application (forestry, land cover, geology, and so forth) of Landsat TM images, which of the contrast enhancement methods discussed is optimum? Explain the reasoning for your selections and any trade-offs that might occur.
5. Explain the principle that Fourier Transform is used for noise removal and other appropriate image processing tasks.
6. Why have so many different vegetation indices been developed? Why not use just one? A vegetation index is not necessarily equally useful for all study areas. For a subject of interest to you (land cover, forestry, geology, etc.) evaluate the significance of the use of vegetation indices by citing examples of how these indices might be used. Also, list some situations for which vegetation indices might be less useful.
7. For land cover analysis, atmospheric correction is not as important as that for a hydrologic or geologic analysis. Justify this statement with reasons.
8. In conducting geometric correction, is the distributional pattern of ground control points important? Why is it sometimes necessary to select questionable points in order to obtain a geographically even distribution?
9. Compare and contrast between supervised and unsupervised image classification in terms of advantages and disadvantages. When would one approach be more appropriate than the other?
10. Evaluate the usefulness of image classification in analysis of land cover and forestry.
11. What are some possible effects of very high spatial resolution satellite imagery on image classification? (Hint: consider object-based image classification.)
12. Explain why and how hyperspectral images need to have image processing and information extraction methods different from those for multispectral imagery.

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